



KTU NOTES

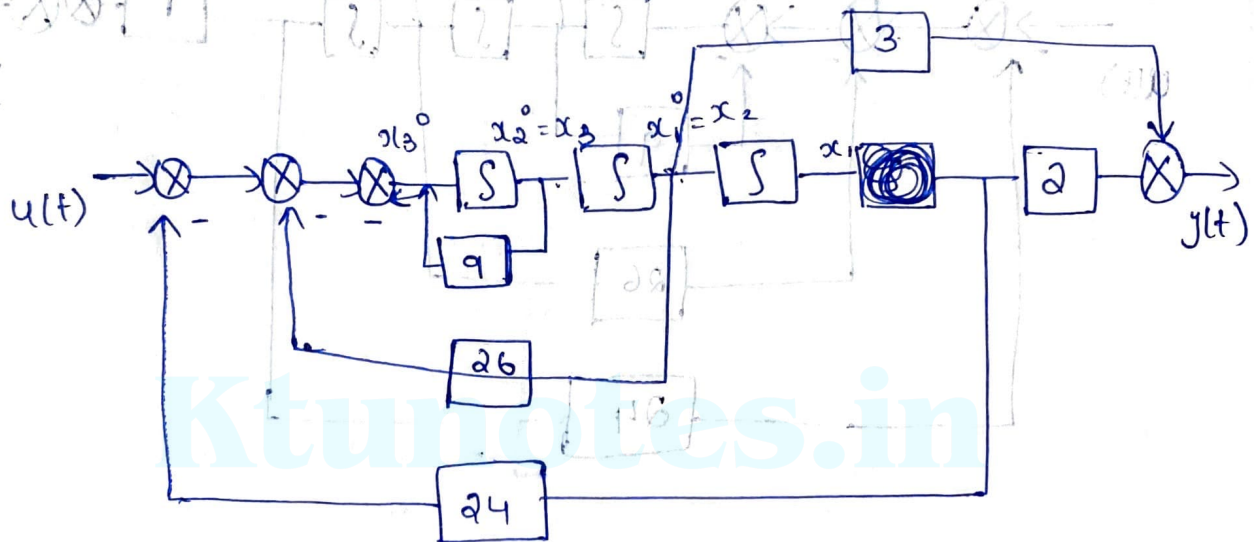
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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

 Website: www.ktunotes.in

(a) Find the phase variable or direct decomposition form of TF?

$$\frac{Y(s)}{U(s)} = \frac{3s+2}{s^3+9s^2+26s+24}$$



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

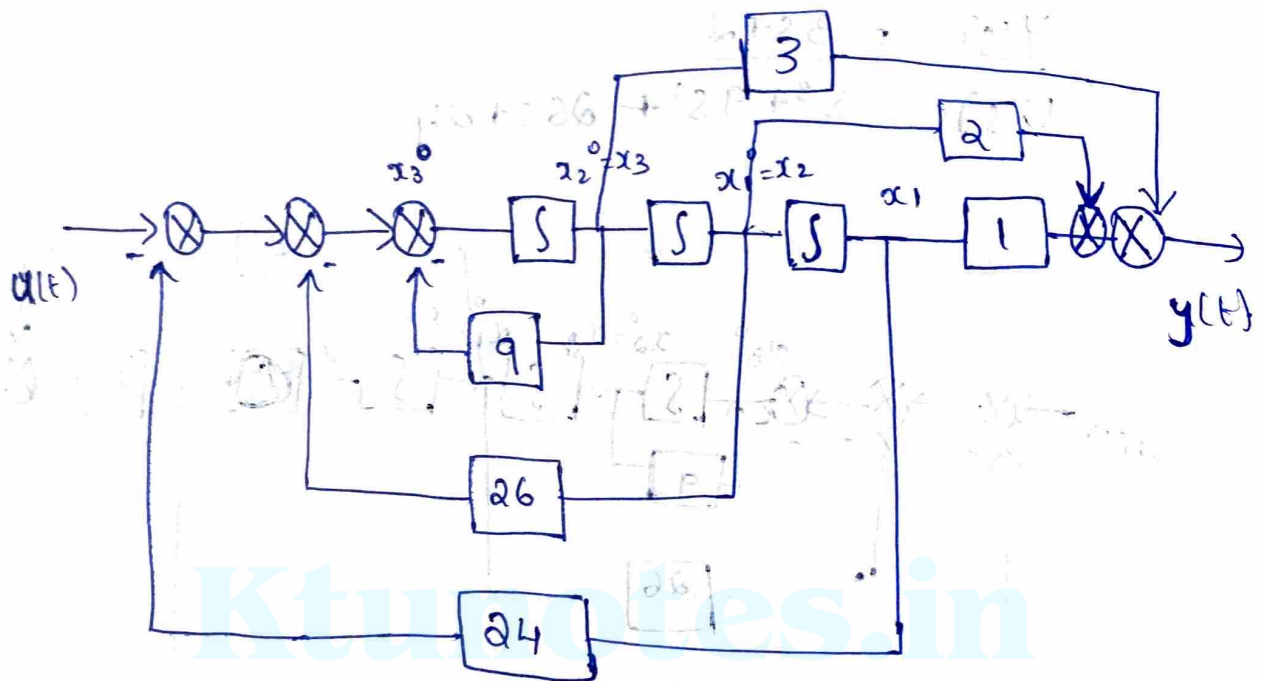
$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + u(t)$$

$$y = 3x_2 + 2x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$(2) \frac{y(s)}{u(s)} = \frac{3s^2 + 2s + 1}{s^3 + 9s^2 + 26s + 24}$$



$$\dot{x}_3 = -9x_3 - 26x_2 - 24x_1 + u(t)$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_1 = x_2$$

$$y = 3x_3 + 2x_2 + 1x_1$$

$$= 3\dot{x}_3 + 2x_2 + x_1$$

$$\begin{bmatrix} x_1^0 \\ x_2^0 \\ x_3^0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} \dot{x} = Ax + B \cdot u \\ y = Cx + D \cdot u \end{cases}$$

(3) Form the differential equation from the state model

$$2y'' + 18y' + 52y + 48 \int_0^t y \, dt = 6u + 4u + 2 \int_0^t u \, dt$$

$$\frac{2d^2y}{dt^2} + 18\frac{dy}{dt} + 52y + 48 \int_0^t y \, dt = \frac{6du}{dt} + 4u + 2 \int_0^t u \, dt$$

$$2s^2 Y(s) + 18s Y(s) + 52Y(s) + 48 \frac{Y(s)}{s}$$

$$= 6s U(s) + 4U(s) +$$

$$2 \frac{U(s)}{s}$$

$$2s^3 Y(s) + 18s^2 Y(s) + 52s Y(s)$$

$$+ 48 Y(s) = 6s^2 U(s) + 4s U(s) + 2U(s)$$

$$= 2s^3 + 18s^2 + 52s + 48 \left[Y(s) \right] = \left[6s^2 + 4s + 2 \right] U(s)$$

$$\frac{y(s)}{u(s)} = \frac{6s^2 + 4s + 2}{2s^3 + 18s^2 + 52s + 48}$$

$$= \frac{3s^2 + 2s + 1}{s^3 + 9s^2 + 26s + 24}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$(1) \frac{y}{s} = \frac{(1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s}}{s}$$

$$(1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s} =$$

$$(1) \frac{y}{s}$$

$$(1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s}$$

$$(1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s} = (1) \frac{y}{s} +$$

$$(1) \frac{y}{s} +$$

$$(1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s} = (1) \frac{y}{s} + (1) \frac{y}{s} + (1) \frac{y}{s}$$

(4) Find the canonical or diagonal form or parallel

form of

$$\frac{Y(s)}{U(s)} = \frac{2}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

$$2 = A(s+3) + B(s+2)$$

$$s = -3$$

$$s = -2$$

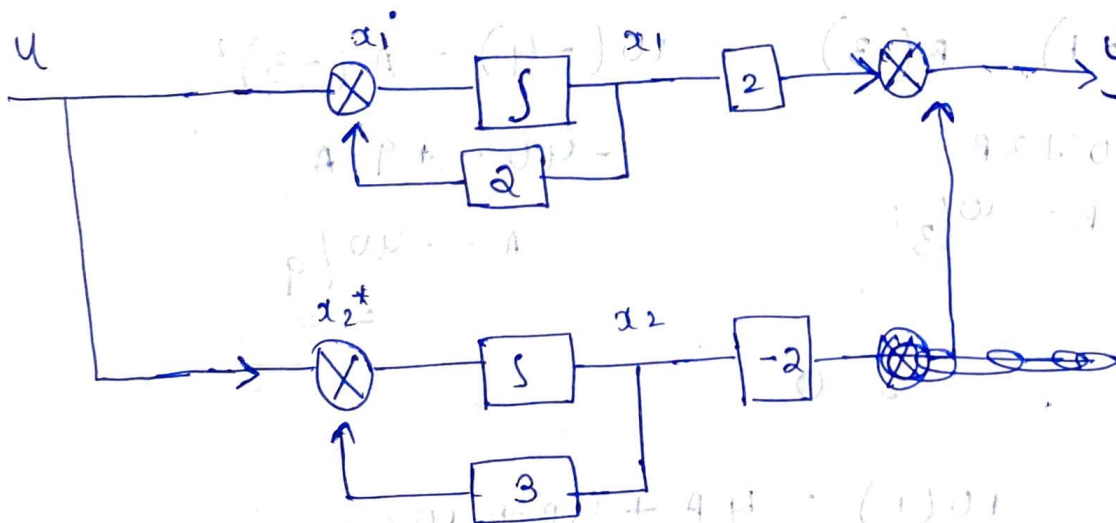
$$2 = B(-1)$$

$$2 = A(1)$$

$$B = -2$$

$$A = 2$$

$$\therefore \frac{Y(s)}{U(s)} = \frac{2}{(s+2)} + \frac{-2}{s+3}$$



$$\dot{x}_1 = -2x_1 + u$$

$$\dot{x}_2 = -3x_2 + u$$

$$y = 2x_1 + (-2x_2)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(5) Jordan Canonical

$$\frac{Y(s)}{U(s)} = \frac{10(s+1)}{(s+5)(s+2)^2}$$

$$\frac{10(s+1)}{(s+5)(s+2)^2} = \frac{A}{(s+5)} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2}$$

$$10(s+1) = A(s+2)^2 + B(s+5) + C(s+5)(s+2)$$

$$s = -2$$

$$s = -5$$

$$10(s+1) = B(s+5)$$

$$10(s+1) = A(s+2)^2$$

$$10(-1) = B(3)$$

$$10(-4) = A(-3)^2$$

$$-10 = 3B$$

$$-40 = 9A$$

$$B = -10/3$$

$$A = -40/9$$

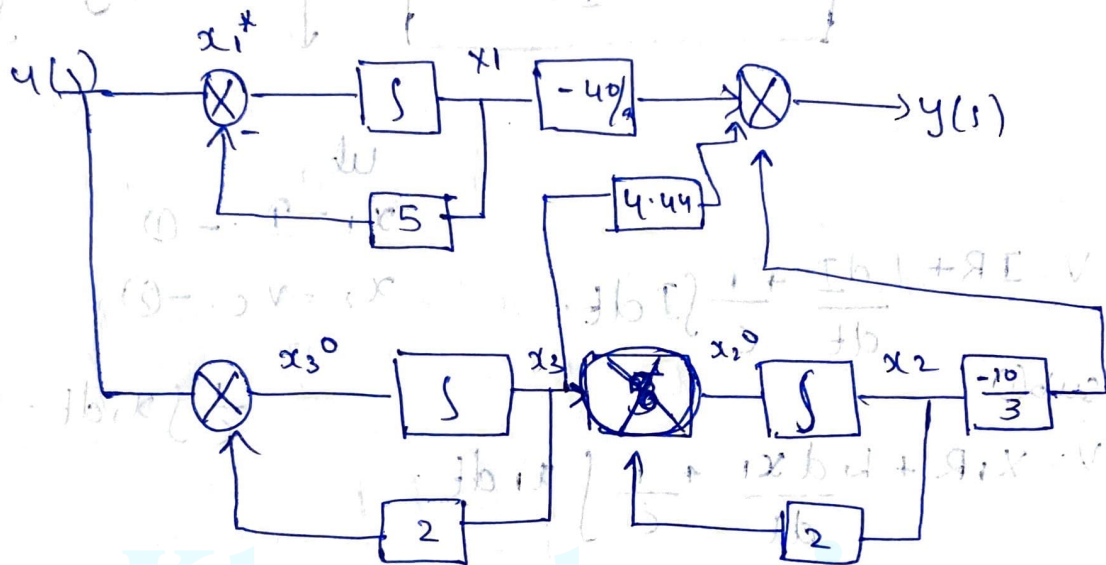
$$s = 0$$

$$10(1) = 4A + 5B + 10C$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & s \\ s & 0 \end{bmatrix} C = \begin{bmatrix} 4 \\ 5 \\ 10 \end{bmatrix}$$

$$\therefore \frac{y(s)}{u(s)} = \frac{-40/9}{(s+5)} + \frac{-10/3}{(s+2)^2} + \frac{4.44}{(s+2)}$$

$s^2 + 4s + 4$



$$\dot{x}_1 = -5x_1 + u$$

$$\dot{x}_2 = -2x_2 + x_3$$

$$\dot{x}_3 = -2x_3 + u$$

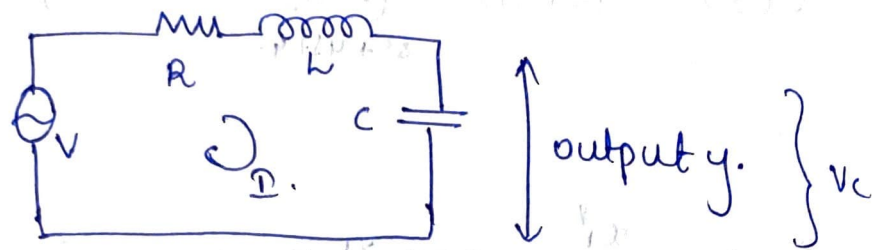
$$y = \frac{-40}{9} x_1 + \frac{-10}{3} x_2 + \frac{40}{9} x_3$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -5 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} \frac{-40}{9} & \frac{-10}{3} & \frac{40}{9} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(6) Find the state model of series RLC circuit.



$$\text{Let, } x_1 = I \quad \text{--- (1)}$$

$$x_2 = v_c \quad \text{--- (2)}$$

$$V = IR + L \frac{dI}{dt} + \frac{1}{C} \int I dt$$

sub (1)

$$V = x_1 R + L \frac{dx_1}{dt} + \frac{1}{C} \int x_1 dt$$

$$\frac{1}{C} \int x_1 dt = v_c$$

$$V = x_1 R + L x_1' + x_2$$

$$L x_1' = -x_1 R - x_2 + V$$

$$x_1' = -\frac{x_1 R}{L} - \frac{1}{L} x_2 + \frac{V}{L}$$

$$x_1' = -x_1 \frac{R_0}{L} - \frac{1}{L} x_2 + \frac{V}{L}$$

$$v_c = \frac{1}{C} \int x_1 dt$$

$$x_2 = \frac{1}{C} \int x_1 dt$$

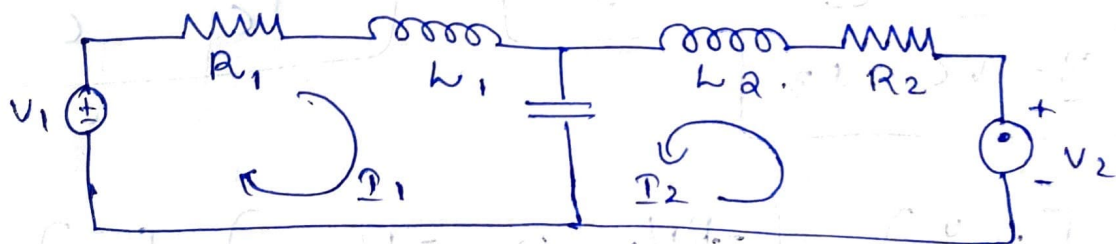
$$x_2' = \frac{1}{C} x_1$$

$$\text{output } y = v_c = x_2$$

$$\therefore \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} V$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(7) Obtain state space model. Take V_c as output variable.



$$x_1 = I_1$$

$$x_2 = I_2$$

$$x_3 = V_c$$

$$V_1 = I_1 R_1 + L_1 \frac{dI_1}{dt} + \frac{1}{C} \int (I_1 + I_2) dt$$

$$V_1 = x_1 R_1 + L_1 \frac{dx_1}{dt} + x_3$$

$$L_1 x_1' = V_1 - x_1 R_1 - x_3$$

$$\dot{x}_1 = -\frac{R_1}{L} x_1 - \frac{x_3}{L_1} + \frac{1}{L_1} v_1$$

$$v_2 = \mathcal{I}_2 R_2 + L_2 \frac{d\mathcal{I}_2}{dt} + \frac{1}{C} \int (\mathcal{I}_2 + \mathcal{I}_1) dt$$

$$v_2 = x_2 R_2 + L_2 \dot{x}_2 + x_3$$

$$\dot{x}_2 = -\frac{R_2}{L_2} x_2 - \frac{1}{L_2} x_3 + \frac{1}{L_2} v_2$$

$$v_C = \frac{1}{C} \int (\mathcal{I}_1 + \mathcal{I}_2) dt$$

$$x_3 = \frac{1}{C} \int (\mathcal{I}_1 + \mathcal{I}_2) dt$$

$$\dot{x}_3 = \frac{1}{C} (x_1 + x_2)$$

$$\Rightarrow \dot{x}_3 = \left(\frac{1}{C} x_1 + \frac{1}{C} x_2 \right)$$

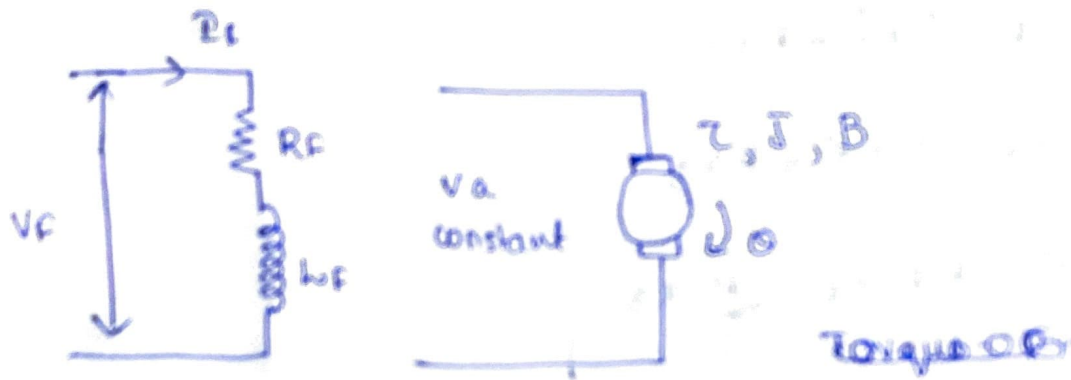
$$y = v_C = x_3$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -R_1/L & 0 & -1/L_1 \\ 0 & -R_2/L_2 & -1/L_2 \\ 1/C & 1/C & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/L_1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$+ \begin{bmatrix} 1/L_1 & 0 \\ 0 & 1/L_2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(8) Determine the state model of field controlled DC motor?



$$\tau = \frac{J d^2 \theta}{dt^2} + B \frac{d\theta}{dt}$$

here torque $\tau \propto I_f$

[field current has field flux]

$$V_f = I_f R_f + L_f \frac{dI_f}{dt}$$

$$I_f = x_1$$

$$V_f = u$$

$$V_f = x_1 R_f + L_f \frac{dx_1}{dt}$$

$$V_f = x_1 R_f + L_f \dot{x}_1$$

$$\dot{x}_1 = \frac{V_f}{L_f} - \frac{R_f}{L_f} x_1$$

$$\dot{x}_1 = \frac{u - 1}{L_f} - \frac{R_f}{L_f} x_1$$

$$\tau \propto I_f$$

$$\tau = k_f I_f$$

$$\tau = \frac{J d^2 \theta}{dt^2} + B \frac{d\theta}{dt}$$

$$\text{let } \theta = x_2$$

$$0 = x_2$$

$$\frac{dx_2}{dt} = x_3$$

$$x_2 = x_3$$

$$K_F \cdot I_F = J \dot{x}_3 + B x_3$$

$$K_F x_1 = J \dot{x}_3 + B x_3$$

$$\dot{x}_3 = \frac{K_F}{J} x_1 - \frac{B}{J} x_3$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{R_F}{L_F} & 0 & 0 \\ \frac{1}{L_F} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_F} \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$